

Grading Rubric: Unmet Clinical Need and Robotic Solution

Medical Robotics — Prof. Marilena Vendittelli

Total: 100 points. Each criterion is scored on a four-level scale and weighted as shown.

1. Clinical Need: Definition and Quantification 15 points

Excellent (13–15)

Problem stated precisely. Population, incidence, complication or failure rate, and downstream consequences quantified with current data from authoritative sources. Reader finishes the section knowing exactly why this matters and at what scale.

Good (10–12)

Problem clearly defined and quantified on at least two dimensions. Sources are appropriate but coverage is uneven.

Adequate (6–9)

Problem identified but quantification is partial, dated, or relies on weak sources. Some claims unsupported.

Weak (0–5)

Problem stated in general terms. Little or no quantification. Reader cannot tell whether this affects hundreds or millions of patients.

2. Analysis of Current Standard of Care 15 points

Excellent (13–15)

Standard of care described accurately and concretely. Existing devices or robotic systems, if any, identified by name with a specific account of what they do and what they fail to do. The gap addressed by the proposed solution is unambiguous.

Good (10–12)

Standard of care described. Gaps identified but stated somewhat generally.

Adequate (6–9)

Standard of care described superficially. Existing solutions mentioned but not critically analyzed. Claims of inadequacy are unsubstantiated.

Weak (0–5)

Misrepresents or misunderstands current practice. Ignores existing solutions that obviously apply. Reader cannot trust the clinical framing.

3. Identification of the Human Limitation

10 points

Excellent (9–10)

A specific human limitation is named (precision, tremor, radiation, fatigue, force control, presence, consistency, access) and tied directly to the proposed mechanism. The robotic concept clearly *follows from* this limitation.

Good (7–8)

Human limitation named. Connection to the solution is present but underdeveloped.

Adequate (4–6)

Limitation named only generally (“robots are more precise”). Weak connection to the proposed design.

Weak (0–3)

No clear human limitation identified. The case for a robot is asserted rather than argued.

4. Quality of the Proposed Robotic Solution

20 points

Excellent (17–20)

Concept is specific and coherent. Robotic task, kinematic architecture, degrees of freedom, sensing, autonomy level, and human-in-the-loop role are all addressed with clear reasoning. Design choices follow from the clinical and human-factors analysis. The sketch communicates the concept on its own.

Good (13–16)

Concept is clear and reasonable. Most required elements present, some at lower depth. Diagram is functional.

Adequate (8–12)

Concept is generic or under-specified. Some elements missing or asserted without rationale. Reads as a description of an existing platform with a new label.

Weak (0–7)

Concept is vague, internally inconsistent, or technically implausible. Diagram missing, illegible, or uninformative.

5. Feasibility Argument

10 points

Excellent (9–10)

Identifies specific enabling technologies and explains why they make this solution viable now. Acknowledges what is still missing and what would have to be developed versus integrated.

Good (7–8)

Identifies enablers but treats them at a surface level. Distinction between mature and immature components is partial.

Adequate (4–6)

Generic appeal to “advances in AI” or “miniaturization” without specifics.

Weak (0–3)

Feasibility unaddressed or contradicted by stated technical requirements.

6. Regulatory and Reimbursement Pathway**10 points****Excellent (9–10)**

Plausible regulatory class identified with a specific predicate or pathway. Payer identified, with a concrete view on existing or proposed reimbursement. Awareness of who actually buys the device and why.

Good (7–8)

Regulatory and payment discussed but with gaps or minor errors.

Adequate (4–6)

Treated superficially. Common errors: “FDA will approve it because it works,” “it will save money so hospitals will buy it.”

Weak (0–3)

Absent, or fundamentally misunderstands how medical devices reach the market.

7. Risks and Failure Modes**10 points****Excellent (9–10)**

Three substantive risks identified across clinical, technical, and commercial dimensions, each with specific mitigations. Demonstrates honest engagement with what could go wrong.

Good (7–8)

Risks identified but skewed to one dimension, or mitigations are generic.

Adequate (4–6)

Risks listed but minimized or treated as solvable in passing.

Weak (0–3)

Risks omitted or framed only as opportunities. No genuine self-critique.

8. Sources and Evidence

10 points

Excellent (9–10)

10+ peer-reviewed sources, well-chosen and accurately represented. Claims traceable to citations. Up-to-date.

Good (7–8)

Meets the source minimum. Mix of strong and weaker sources. Most claims supported.

Adequate (4–6)

Meets the minimum but leans on review articles, dated sources, or vendor materials. Some claims unsupported.

Weak (0–3)

Below the source minimum, and sources misrepresented or misattributed.

Cross-Cutting Penalties

- **Up to –10 points: Solution does not need a robot.** If a sensor, drug, workflow change, or training intervention would address the problem at lower cost and equal effect, and the report does not engage with that possibility, expect significant deduction.
- **Up to –5 points: Failure to distinguish “no robot exists” from “current robot is imperfect.”** Both are valid framings, but they require different arguments. Mixing them up weakens the analysis.
- **Up to –5 points: Reinventing an existing product without acknowledging it.** Da Vinci, Mako, Ion, CorPath, CyberKnife, and similar are well known. Proposing them under a new name will be penalized; building on them with a clearly novel contribution will not.
- **Up to –5 points: Writing and presentation.** Disorganized structure, careless figures, or sloppy referencing.

Bonus

- **Up to +5 points: Original primary research.** Direct conversation with a clinician practicing in the relevant area, observation of the procedure, or quantitative analysis the student performed themselves. Document it.